

Daniel Casadevall Jauhiainen

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→ **Pipeline Technical Director** and **R&D-focused** developer in VFX-Animation, with a strong dual background in software development and **technical art**. Currently pursuing Animation & VFX with a self-directed focus on generative AI, geometry processing, and pipeline automation; aiming toward graduate research (MSc/PhD) in computer graphics or applied ML

SELECTED PROJECTS

I. Project Minivita - Interactive AI Character Framework September 2025

Developed a real-time **multimodal human representation** system integrating an interactive **SMPL-X motion diffusion** framework with an embeded vector database for dynamic character memory and centralized **LLM** execution. Optimized and quantized to run efficiently within consumer GPU memory limits. Rendered via **PySide OpenGL** viewport utilizing a **USD-based** geometry pipeline, implementing an asynchronous **96-frame** chunk system allowing low-latency **autonomous** digital character behavior.

II. AutoMayTex - AI-Driven Texturing Plug-in for Autodesk Maya March 2026 - Ongoing

Developed an **automated** diffusion texturing pipeline using **Maya API** running on a consumer GPU. Capturing multi-angle tetrahedral renders, and processing them via a local **FastAPI** server running **SDXL ControlNet**. Orchestrated via a **PySide6** GUI, it isolates the heavy runtime in a dedicated Python venv.

III. USD Pipeline Lead - ENCORE shortfilm Production - Prism September 2025 - Ongoing

Developed custom **Python** integration to embed **Pixar USD** asset variants and references seamlessly into the **Prism Pipeline** architecture. Deployed as the core pipeline for ENCORE (shortfilm, in production), connecting Houdini and Maya using Prism base code across all departments for a team of ~16 students.

IV. Procedural Generation Engine - Houdini Digital Assets (HDAs) 2025

Designed and implemented custom Houdini Digital Assets (HDAs) for **procedural** generation of urban architecture and complex **plant geometry**. Focused on building scalable team-node networks and exposed parameters tailored for layout artist integration.

V. tLoud Studio Manager - Production Management Web App February 2026 - Ongoing

Web-based asset task management platform for team production **workflows**. Asynchronous Python backend with a **Excel-driven** and **Discord** backend. Dynamic task assignments and multi-user progress monitoring. The project was made for organization of a **16 people** human team to work in ENCORE shortfilm project.

VI. Evolutive Biological Simulation - Neural Behavior Simulation 2022

Engineered a 2D biological simulation framework in Python to calculate **multi-agent** cell adaptation and behavioral inheritance across thousands of generations. Designed a **genetic pipeline** that compiled serialized **28-bit binary genomes** directly into modular sensory-input and motor-output execution loops. Coded entirely using the Python standard library.

EDUCATION

2023-2027	Degree in Animation & VFX (In Progress) La Salle Bonanova University · Barcelona
2024	Computer Science & Software Engineering (Paused) Universitat Oberta de Catalunya (UOC) · Barcelona
2022	Technological Baccalaureate - Second Year La Sedeta Institute · Barcelona
2021	High School (Lukio) - First Year Etelä-Tapiolan lukio · Helsinki

LANGUAGES

Catalan	Native · C2
Spanish	Native · C2
English	Upper · B2
Finnish	Native · C1

LANGUAGES & TECHNOLOGIES

HTML / CSS

JAVASCRIPT

PYTHON

SQL

C++

C#

C

VEX

PHP

PIPELINE & APIS

PIXAR USD	MAYA API
PYSIDE6 / PYQT	FASTAPI
PRISM PIPELINE	PYTORCH
TRANSFORMERS	DIFFUSERS

SOFTWARE

AUTODESK MAYA

SIDEFX HOUDINI

BLENDER

NUKE

UNITY

GIT

LINUX

REASERCH INTERESTS

generative diffusion models for texture and geometry synthesis. Real-time motion and simulated character behavior systems, efficient on-device/consumer-GPU inference for production pipelines. Underlying interest in the boundary between biological and digital life (what it means for a system to behave, adapt, or 'live'.)

ABOUT ME

Autonomous, self-directed learner: Most of the technical skills above were acquired independently, outside formal coursework, driven by wanting to build systems that don't exist yet.

ADDITIONAL EXPERIENCE

Youth Leader (Monitor de Lleure)
Agrupament Lluïsos de Gràcia (MEG)

Managed group dynamics, conflict resolution, and teamwork activities for youth education. Officially certified as a group coordinator.